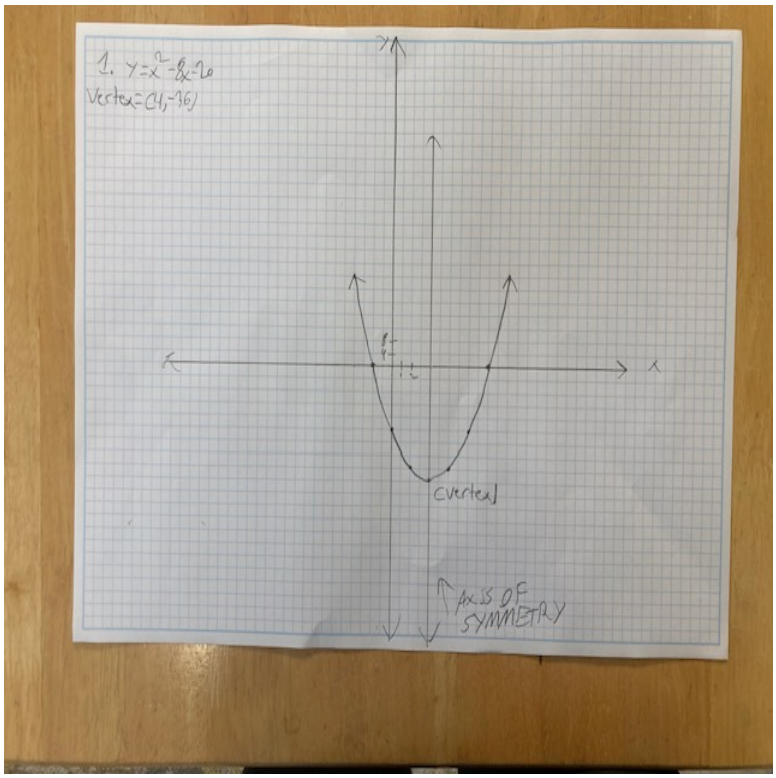
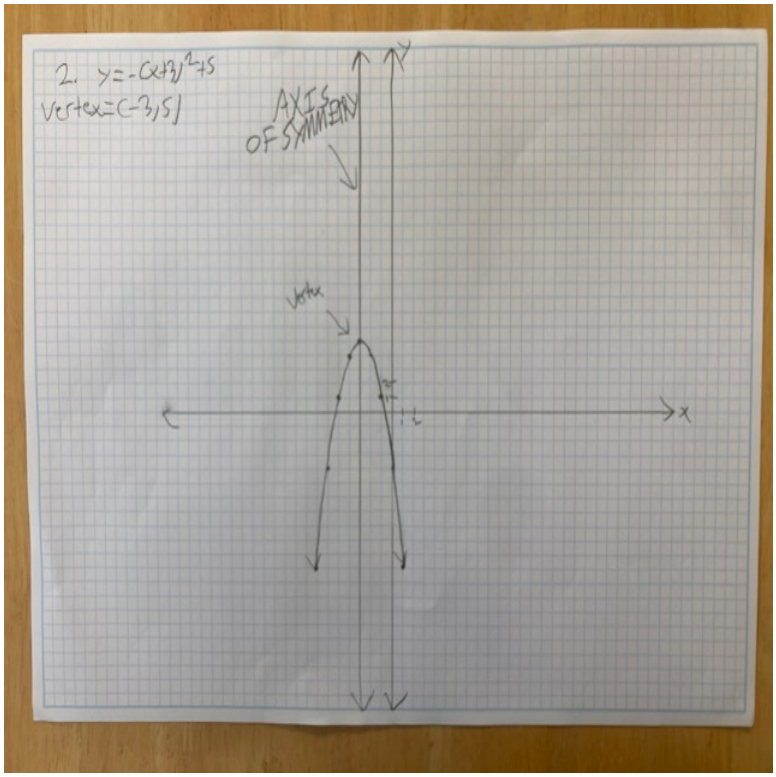
Chapter Test (pg. 81)

October 12, 2023

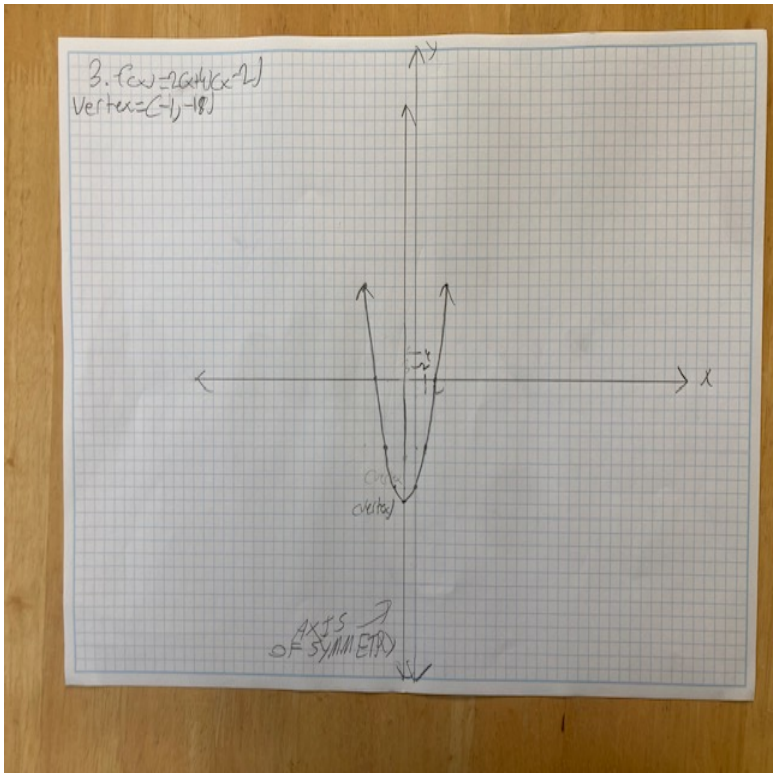
1



2



3



4

$x^2 - 11x + 30$
$\rightarrow (x-6)(x-5)$

5

$z^2 + 2z - 15$
$\rightarrow (z+5)(z-3)$

6

$n^2 - 64$
$\rightarrow (n + 8)(n - 8)$

7

$2s^2 + 7s - 15$
$\rightarrow (2s - 3)(s + 5)$

8

$9x^2 + 30x + 25$
$\rightarrow (3x)^2 + 30x + (5)^2$
$\rightarrow (3x + 5)^2$

9

$6t^2 + 23t + 20$
$\rightarrow (3t + 4)(2t + 5)$

10

$x^2 - 3x - 40 = 0$
$\rightarrow (x - 8)(x + 5) = 0$ (Solve by factoring)
$\rightarrow x = 8 \vee x = -5$ (Use Zero Product Property)

11

$r^2 - 13r + 42 = 0$
$\rightarrow (r - 7)(r - 6) = 0$ (Solve by factoring)
$\rightarrow r = 7 \vee r = 6$ (Use Zero Product Property)

12

$2w^2 + 13w - 7 = 0$
$\rightarrow (2w - 1)(w + 7) = 0$ (Solve by factoring)
$\rightarrow w = \frac{1}{2} \vee w = -7$ (Use Zero Product Property)

13

$10y^2 + 11y - 6 = 0$
$\rightarrow (5y - 2)(2y + 3) = 0$ Solve by factoring)
$\rightarrow y = \frac{2}{5} \vee y = -\frac{3}{2}$ (Use Zero Product Property)

14

$2(m - 7)^2 = 16$
$\rightarrow (m - 7)^2 = 8$
$\rightarrow m - 7 = \pm\sqrt{8}$ (Solve by finding the square root)
$\rightarrow m = 7 \pm \sqrt{8}$
$\rightarrow m = 7 \pm 2\sqrt{2}$ (Simplify the radical)

15

$(x + 2)^2 - 12 = 36$
$\rightarrow (x + 2)^2 = 48$
$\rightarrow x + 2 = \pm\sqrt{48}$ (Solve by finding the square root)
$\rightarrow x = -2 \pm \sqrt{48}$
$\rightarrow x = -2 \pm 4\sqrt{3}$ (Simplify the radical)

16

$(3 + 4i) - (2 - 5i) =$
$\rightarrow (3 - 2) + (4i + 5i) = 1 + 9i$

17

$(2-7i)(1+2i) =$
$\rightarrow 2-7i+4i-14i^2 = 16-3i$

18

$\frac{3+i}{2-3i} =$
$\rightarrow \frac{(3+i)(2+3i)}{(2-3i)(2+3i)} =$
$\rightarrow \frac{6+9i+2i+3i^2}{4+6i-6i-9i^2} =$
$\rightarrow \frac{6+11i+3i^2}{4-9i^2} =$
$\rightarrow \frac{3+11i}{13} =$
$\rightarrow \frac{3}{13} + \frac{11}{13}i$

19

$x^2+4x-14 = 0$
$\rightarrow x^2+4x = 14$
$\rightarrow x^2+4x+4 = 18$ (Add $(\frac{b}{2})^2$)
$\rightarrow (x+2)^2 = 18$
$\rightarrow x+2 = \pm\sqrt{18}$
$\rightarrow x = -2 \pm \sqrt{18}$
$\rightarrow x = -2 \pm 3\sqrt{2}$

20

$x^2-10x-7 = 0$
$\rightarrow x^2-10x = 7$
$\rightarrow x^2-10x+25 = 32$ (Add $(\frac{b}{2})^2$)
$\rightarrow (x-5)^2 = 32$
$\rightarrow x-5 \pm \sqrt{32}$
$\rightarrow x = 5 \pm \sqrt{32}$
$\rightarrow x = 5 \pm 4\sqrt{2}$

21

$4x^2 + 8x + 3 = 0$
$\rightarrow 4(x^2 + 2x) = -3$
$\rightarrow 4(x^2 + 2x + 1) = 1$ (Add $(\frac{b}{2})^2 \times 4$)
$\rightarrow 4(x+1)^2 = 1$
$\rightarrow (x+1)^2 = \frac{1}{4}$
$\rightarrow x + 1 = \pm\sqrt{\frac{1}{4}}$
$\rightarrow x = -1 \pm \frac{1}{2}$
$\rightarrow x = -\frac{1}{2} \vee x = -\frac{3}{2}$

22

$3x^2 + 10x - 5 = 0$
$\rightarrow x = \frac{-10 \pm \sqrt{10^2 - 4(3)(-5)}}{2(3)}$
$\rightarrow x = \frac{-10 \pm \sqrt{160}}{6}$
$\rightarrow x = \frac{-10 \pm 4\sqrt{10}}{6}$
$\rightarrow x = \frac{-5 \pm 2\sqrt{10}}{3}$

23

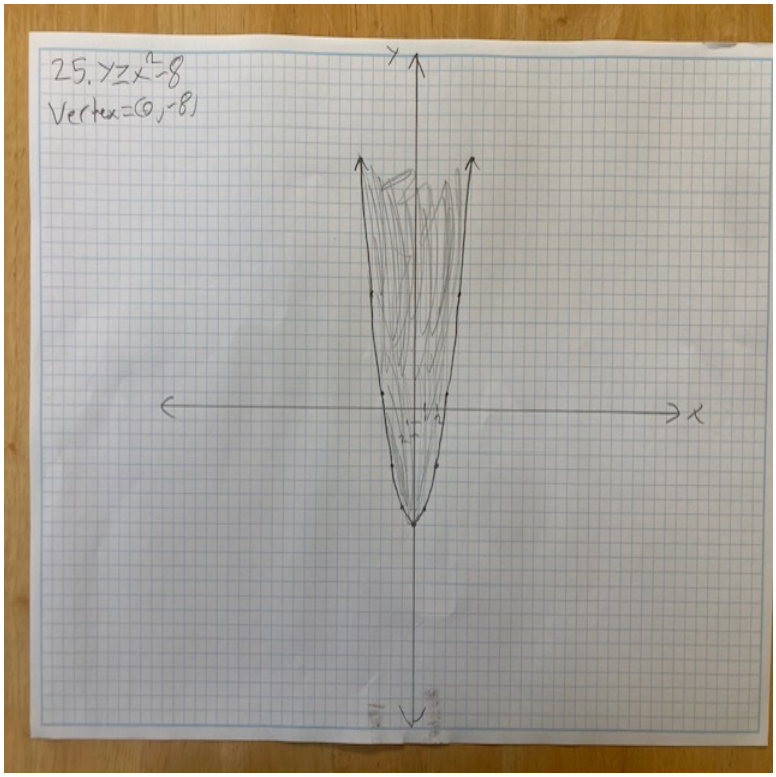
$2x^2 - x + 6 = 0$
$\rightarrow x = \frac{1 \pm \sqrt{(-1)^2 - 4(2)(6)}}{2(2)}$
$\rightarrow x = \frac{1 \pm \sqrt{-47}}{4}$
$\rightarrow x = \frac{1 \pm \sqrt{47}i}{4}$
$\rightarrow x = \frac{1}{4} + \frac{\sqrt{47}}{4}i \vee x = \frac{1}{4} - \frac{\sqrt{47}}{4}i$

24

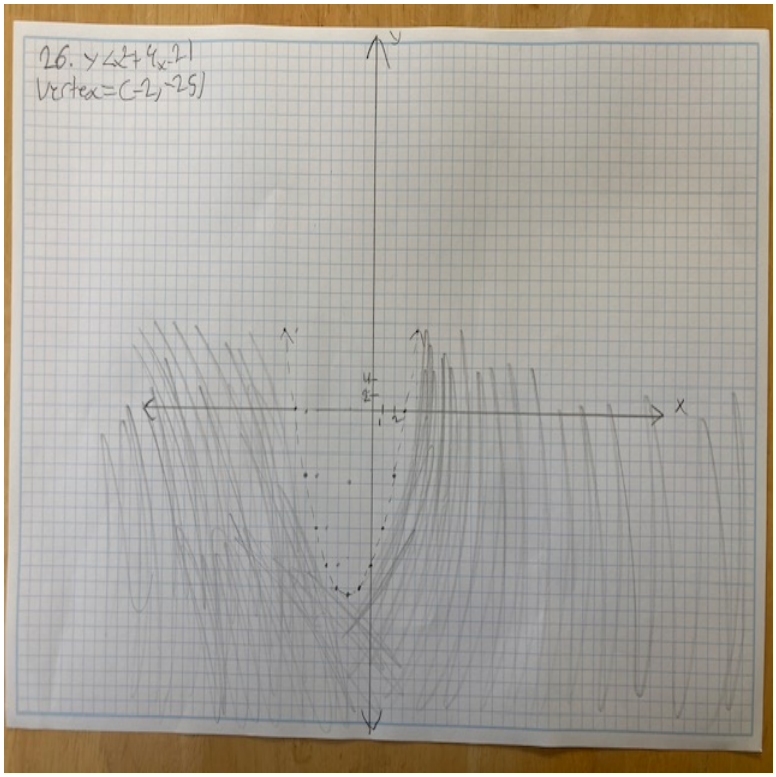
$5x^2 + 2x + 5 = 0$
$\rightarrow x = \frac{-2 \pm \sqrt{2^2 - 4(5)(5)}}{2(5)}$
$\rightarrow x = \frac{-2 \pm \sqrt{-96}}{10}$
$\rightarrow x = \frac{-2 \pm 4i\sqrt{6}}{10}$

$\rightarrow x = \frac{-1 \pm 2i\sqrt{6}}{5}$
$\rightarrow x = -\frac{1}{5} + \frac{2\sqrt{6}}{5}i \vee x = -\frac{1}{5} - \frac{2\sqrt{6}}{5}i$

25

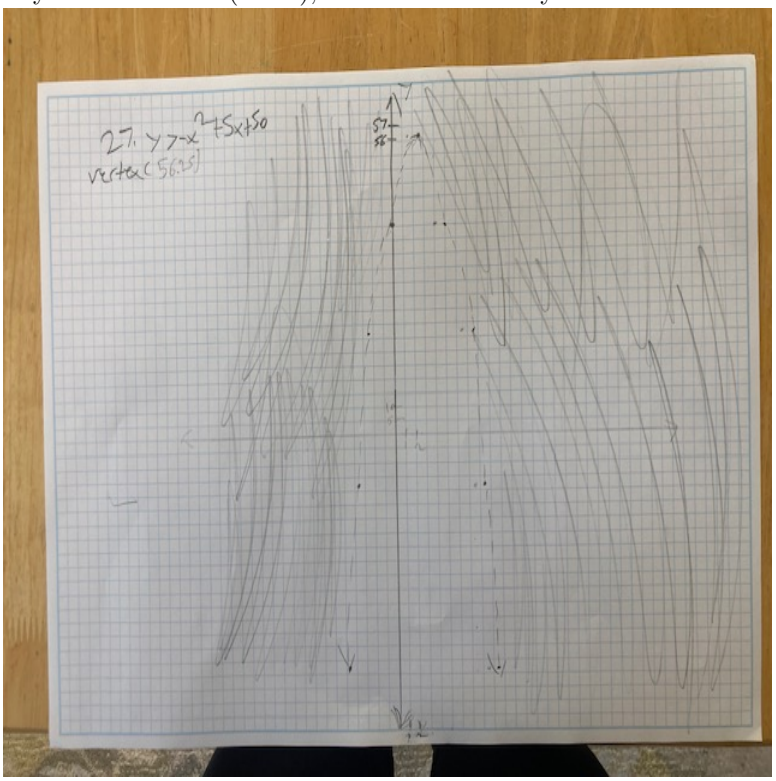


26

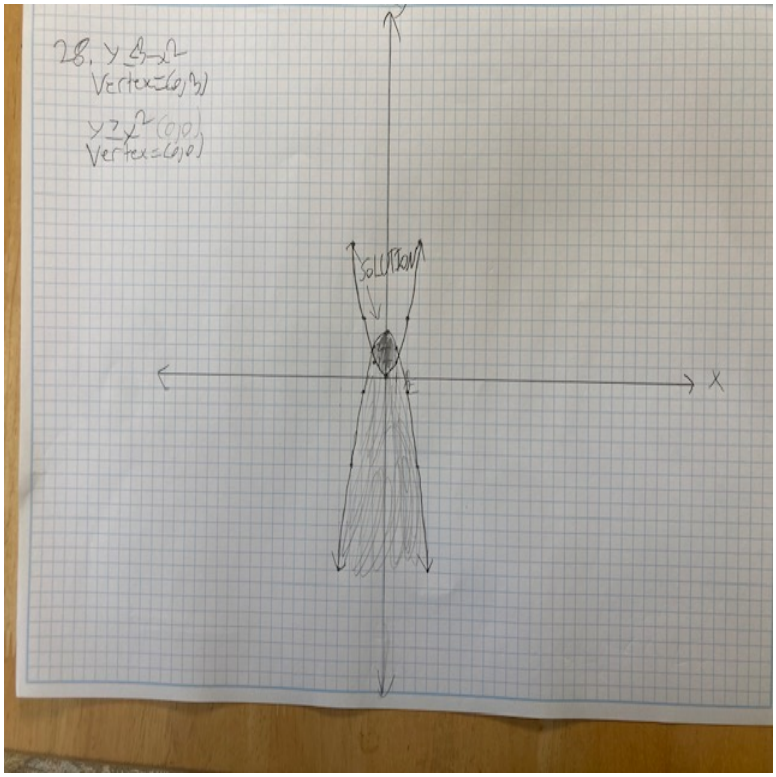


27

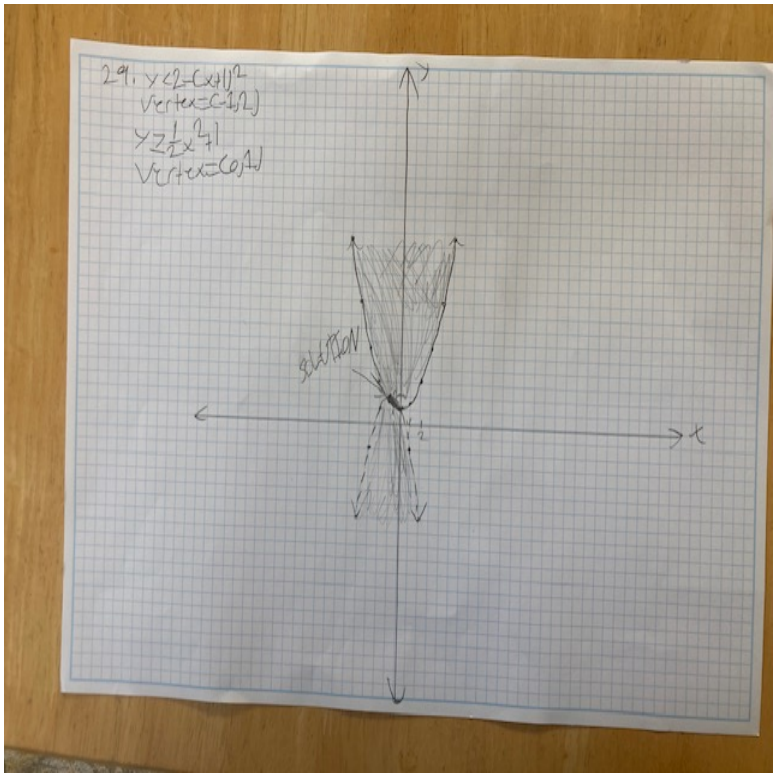
The Picture says the vertex is (56.25), but I meant to say that the vertex is (2.5, 56.25).



28

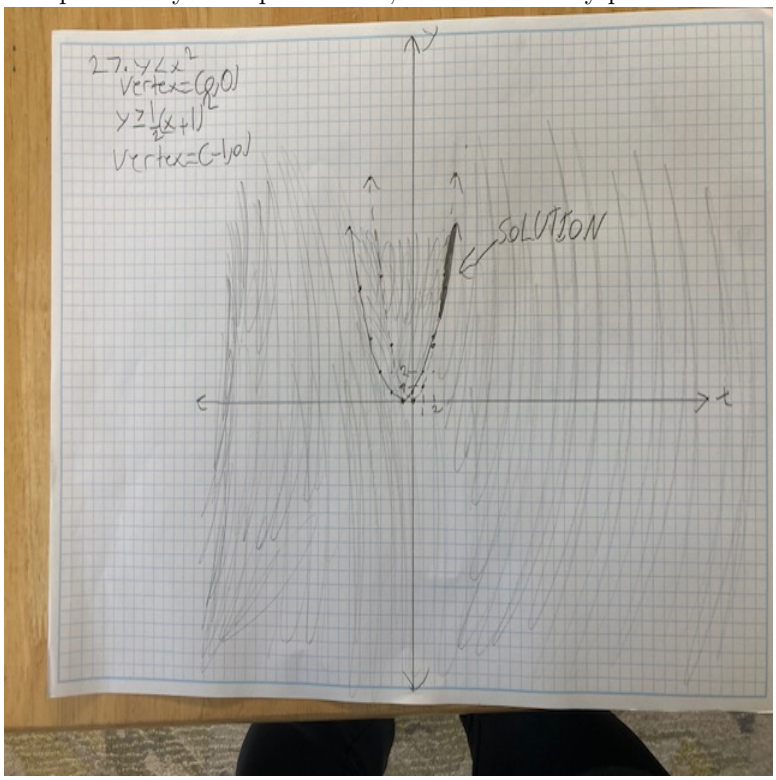


29



30

The picture says it is problem 27, but it is acutally problem 30.



31

By the pythagorean theorem, we know that in a right triangle, $a^2 + b^2 = c^2$. When observing the given diagram, we can tell that $c = 32, a = 9x, b = 16x$. Thus, $32^2 = (9x)^2 + (16x)^2$
$\rightarrow 1024 = 81x^2 + 256x^2$
$\rightarrow 1024 = 337x^2$
$\rightarrow x^2 = \frac{1024}{337}$
$\rightarrow x = \pm\frac{\sqrt{1024}}{\sqrt{337}}$
$\rightarrow x = \pm\frac{\sqrt{1024}\sqrt{337}}{\sqrt{337}\sqrt{337}}$ (Simplify the radical)
$\rightarrow x = \pm\frac{\sqrt{345088}}{337}$
$\rightarrow x = \pm\frac{32\sqrt{337}}{337}$
Because it does not make sense to have x be negative (if this were the case, then

we would have negative measurements), x must be positive, meaning $x = \frac{32\sqrt{337}}{337}$